

Date: 19-May-20	Product Requirements Document (PRD)	
PRD-PFMD	Product1 PRD	

REVISION LOG

REV.#	DATE	AFFECTED SECTNS.	CHANGE DESCRIPTION	AUTHOR/OWNER
N/C	2/18/2020	N/A	Initial Release	Arty Makagon
A	5/12/2020	1	Clarified verbiage pertaining to range configuration options	
		4	Decreased storage temperature range to account for battery requirements	
		5	Specified HDMI and USB versions, increased total number of USB ports to four	
		Appendix 1	added Appendix 1 - Examples of multiple PFMD units configured to work in concert	
		Appendix 2	added Appendix 2 - Example of multiple regions of interest	
		Appendix 3	added Appendix 3 - Relative positional accuracy example	
B	5/18/2020	1	Clarified positional accuracy specification	
		Appendix 1	Stipulated FOV overlap requirement in section iii	
		Appendix 3	Removed	

1 PRODUCT DESCRIPTION / KEY FUNCTIONALITY

Illumination	Onboard IR illumination to allow day or night operation
Run Time	1-week continuous operation, including adequate onboard data storage (upload to cloud for post-processing when network is available)
Active Volume	NOTE: for pictorial descriptions, please see Figure 1 and APPENDIX 1 3m x 3m x RANGE square pyramid frustum, where RANGE is set at the time of unit production as specified by the customer, varying from 3m to 10m - For example: - Configuration A (largest, aka LAB): 3m X 3m X 10m frustum, blind to targets at range less than 5m; see Figure 1A - Configuration B (smallest, aka MINI): 3m X 3m X 3m frustum, blind to targets at range less than 1m; see Figure 1B - For use in larger volumes, multiple units can be calibrated to operate conjointly; see APPENDIX 1
Target Capacity	- Ability to track no fewer than 1,000 non-intersecting targets concurrently - Ability to track multiple intersecting targets (e.g., swarms) concurrently is TBD
Resolution	- Reliably tracks targets with characteristic length of 4mm or greater (i.e., mosquitoes) - Smaller targets are likely to be detected, but lower bound is TBD
Recorded Data	3-D position of each unique target, maintaining track continuity across specified active region(s), see APPENDIX 2 - trajectory uncertainty, with respect to other targets, of ± 2mm along height and width axes and ± 5mm along the range axis - Apparent size estimate of each unique target

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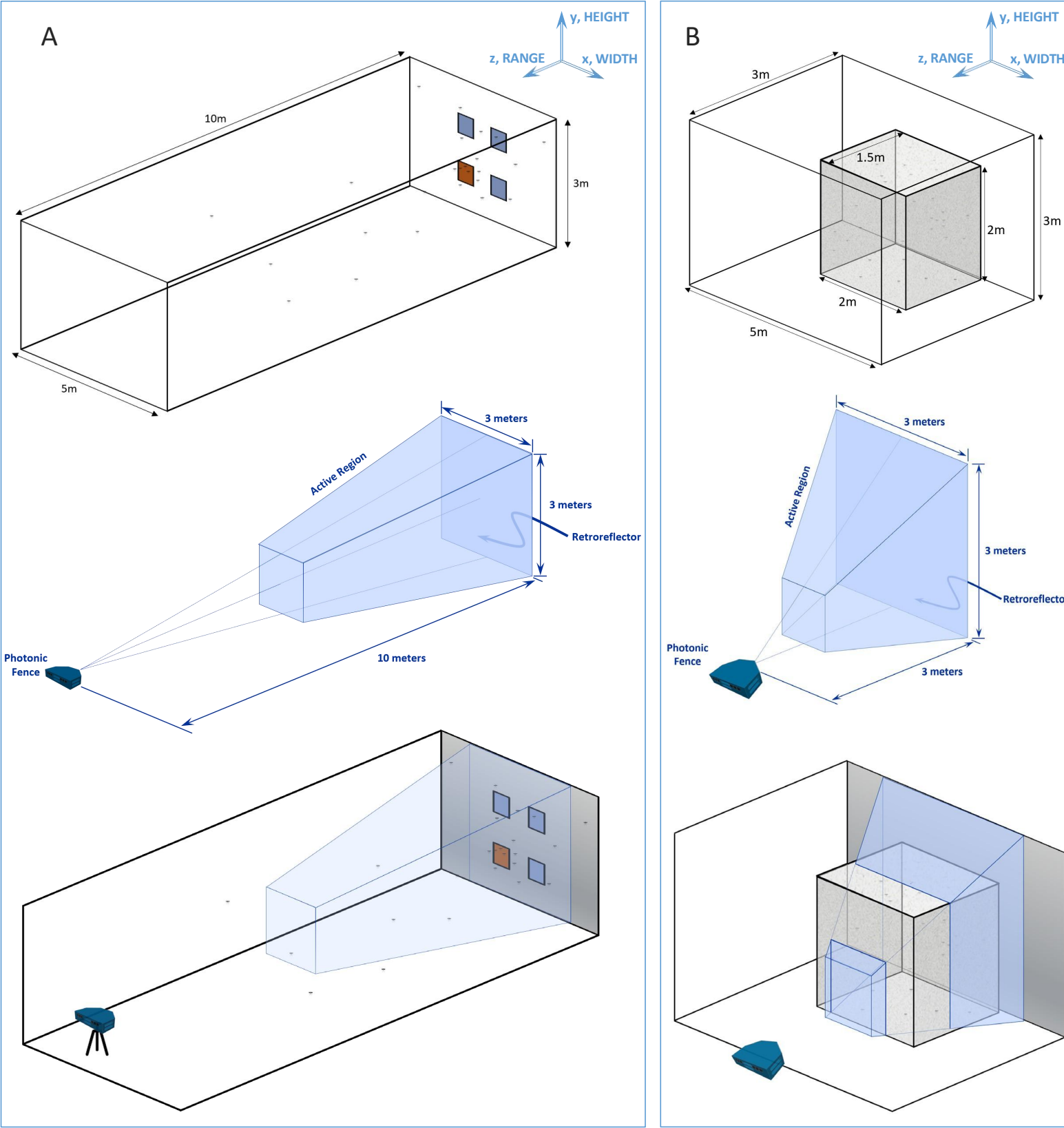


Figure 1: Reference Product1 Installations: A – LAB configuration, B – MINI configuration

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2 STAKEHOLDERS

Target User	- Entomologists and other vector control researchers - Vector control intervention R&D entities
Other	- Bill and Melinda Gates Foundation (BMGF) - Global Good Fund (GG)

3 REGULATORY

Assume that no regulatory constraints exist

Electrical Safety	Package to include UL/FCC certified power supply
Emissions	Compliant with FCC Class A (non-residential) regulations*
Export Control	N/A, but will need analysis to show compliance
Labeling	TBD

* Note: as-built device will need to be tested to confirm compliance

4 OPERATING ENVIRONMENT

Ambient Temperature	- Operational: 5°C - 45°C - Storage: -20°C - 60°C
Sun Exposure	Direct sunlight may raise the operating temperature of the optoelectronic hardware beyond acceptable limits and may also impose excessive optical noise. Installations which would normally expose the PFMD to direct sunlight will need to erect a sunshade.
Moisture Resistance*	Indoor/outdoor, including capability to sustain (but not necessarily operate in) high humidity and rain
Dust Protected*	Protected from limited dust ingress

*IP54 Rated

5 INTERFACES

Hardware	
Electrical Power	12V DC input; 110V - 220 V adapter included in kit 300W power budget; capable of being solar-powered (solar panel, battery, etc. not included) - Integrated battery, allowing device to handle brownouts up to 10 minutes in duration
Network	RJ-45 ethernet port
Human Interface Devices	1 x HDMI (1.4 or newer) port for video 3 x USB-3 ports for peripherals (mouse, keyboard, etc.)
External Storage	1 x USB-3 port
Software	
UX	Onboard: minimal, intended for data-gathering setup and troubleshooting

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	Web app: includes pre-configured dashboards, visualization tools and data export features
File Size	Small, capable of being transferred using a USB thumb drive
Data Processing	- Near-Realtime visualization of key parameters (e.g., # of targets tracked, targets trajectory playback, etc.) - Additional processing accomplished post-data-collection

6 PORTABILITY

Initial Setup and Calibration Routine	Accomplished within 4 hours
Teardown	Accomplished within 2 hours
Mass	[10 kg] max, capable of being moved and hoisted onto tripod by one person

7 PACKAGING

In the Box	- Product1 (PFMD) - Power supply - Ethernet cable - Height-adjustable tripod for erecting the PFMD - Retroreflective screen - Free-standing frame for mounting retroreflective screen
Shipping Crate Impact Requirements	Must withstand drop from 2m height
Labeling	TBD

8 SERVICEABILITY / MAINTAINABILITY

Required Maintenance	- Wipe clean retroreflective backdrop, on demand - Wipe clean front optical surface of the PFMD, on demand
Customer Support System	- Email / phone - Remote troubleshooting capability via ethernet
Returns Process	- Send replacement within 2 weeks and repair returned device - Shipping cost is covered during the warranty period - RMA required
Operability Self-Test	To be executed by customer at customer site. Pass/fail metrics are TBD.

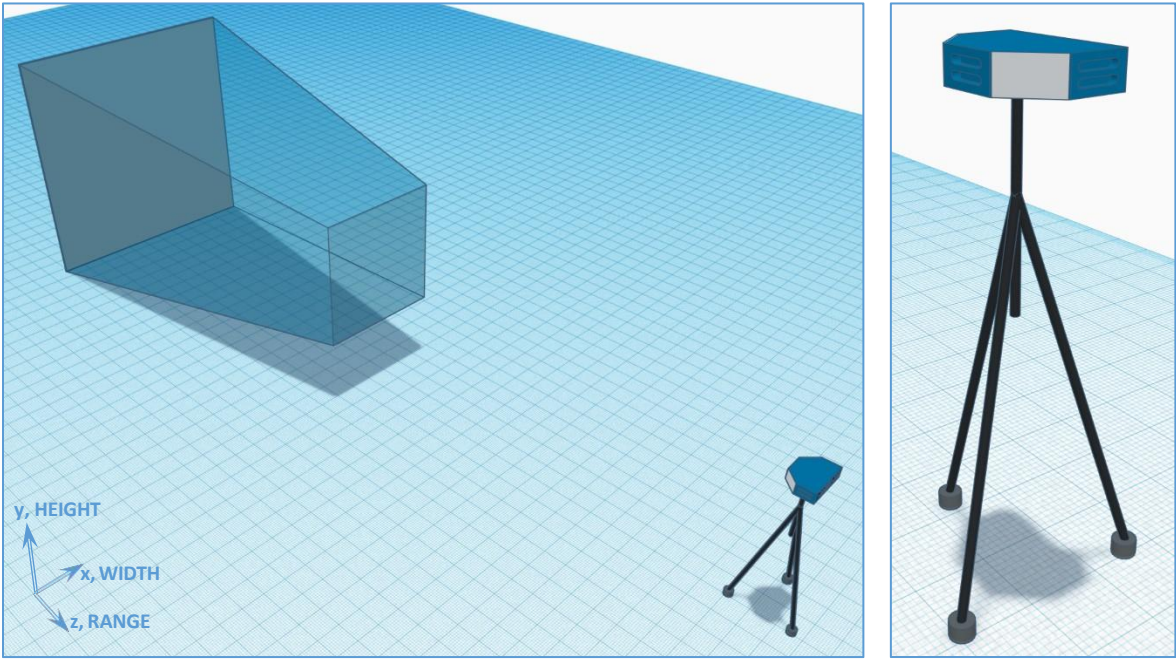
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APPENDIX 1

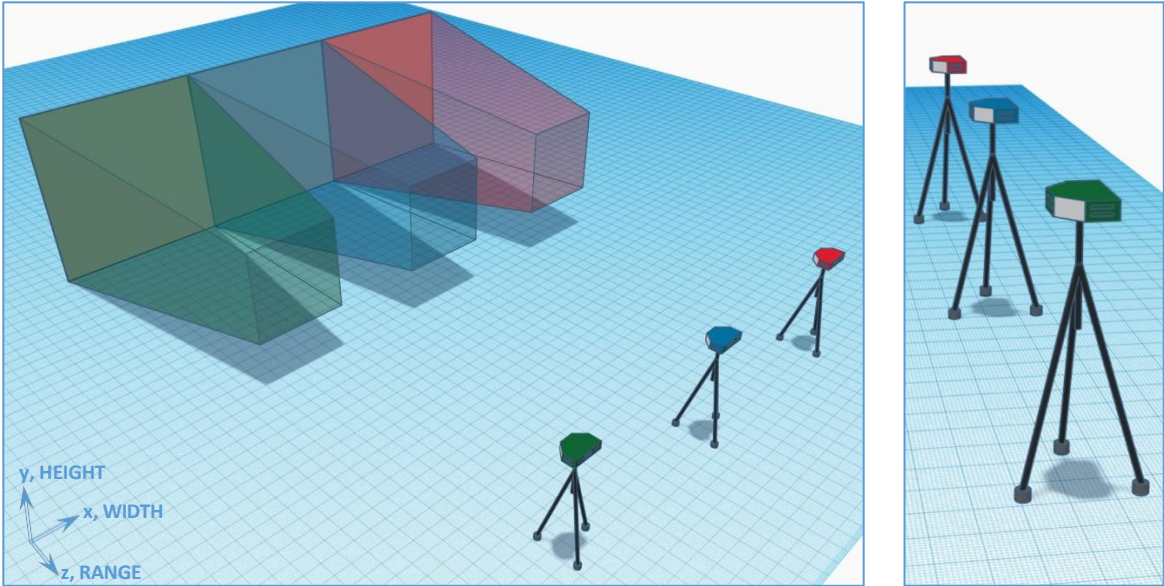
Examples of multiple PFMD units configured to work in concert

A. LAB configuration; for active range dimensions, refer to Figure 1A. PFMD units shown on tripods at 1.5m above ground level for illustration purposes only; units can be positioned at any height to optimize vantage point onto experimental stage.

i. Single unit, shown for reference

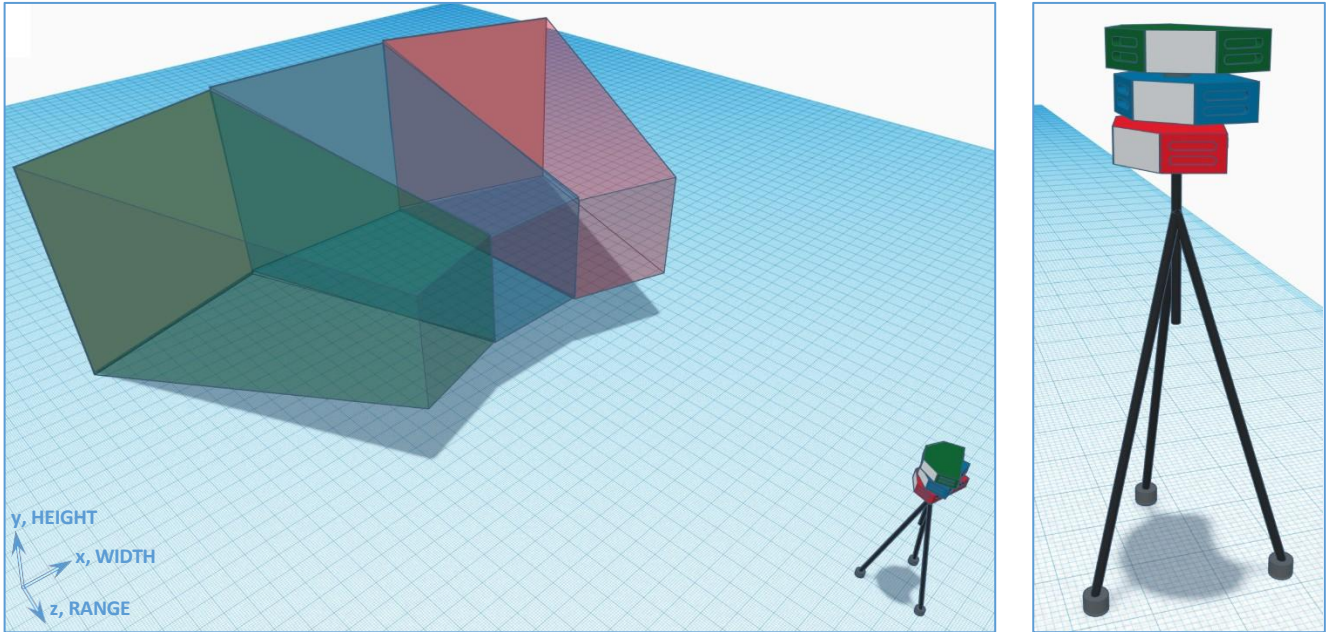


ii. “Array” – multiple units arranged in a linear array for viewing adjacent scenes concurrently

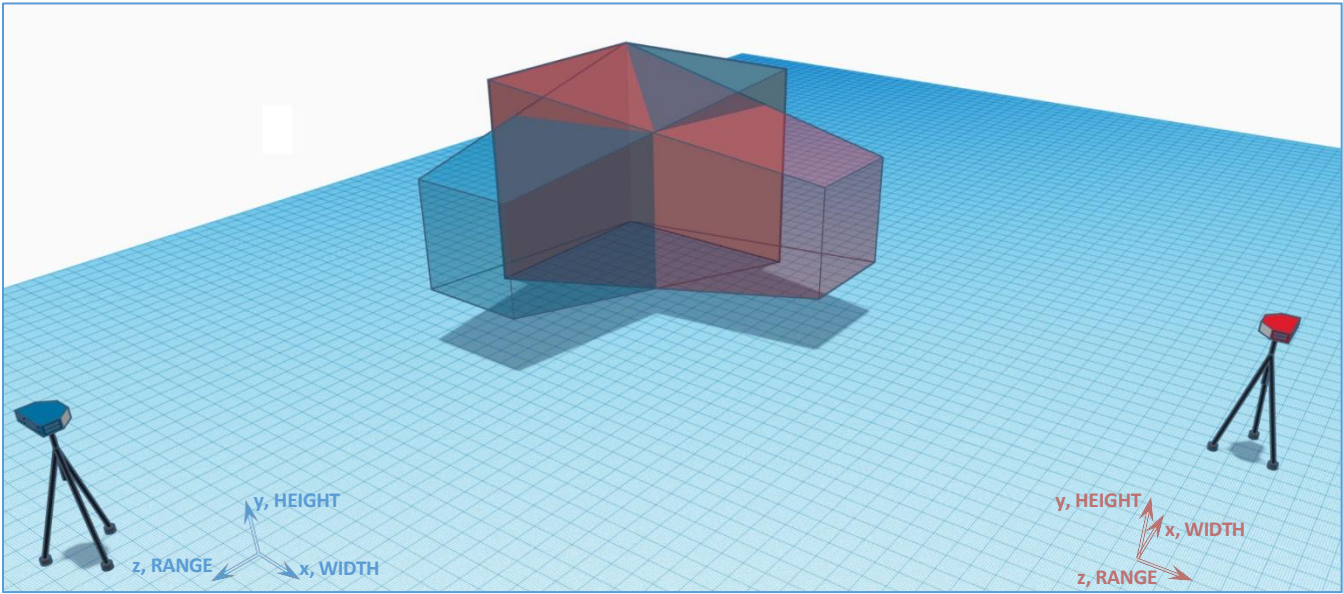


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- iii. “Fan” – multiple units arranged to maximize contiguous active volume. Note that adjacent active regions will need to be configured with adequate overlap to ensure target continuity.



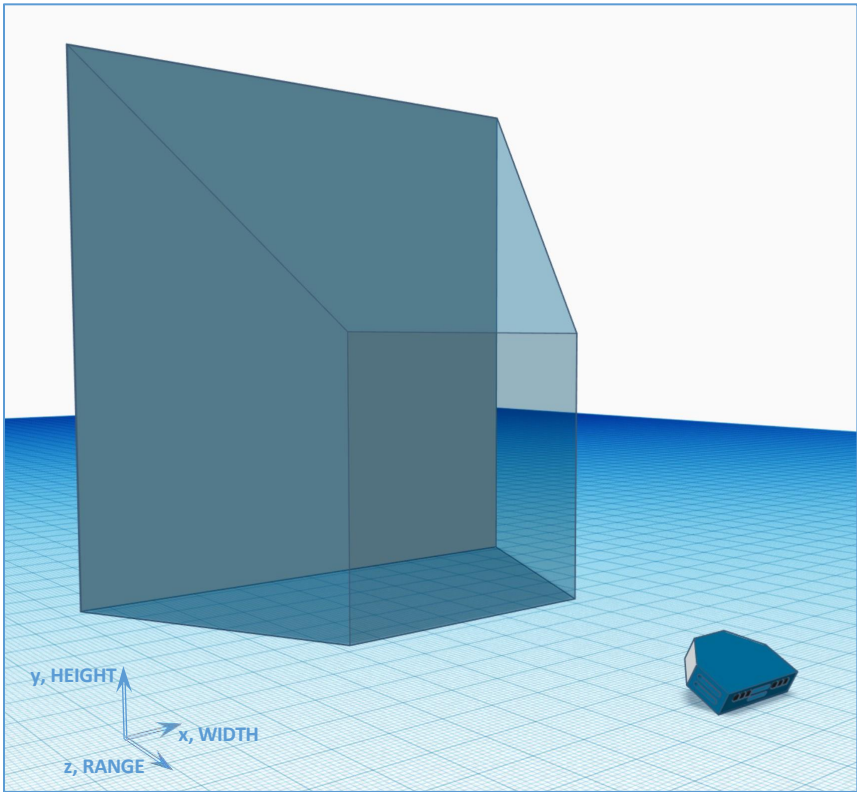
- iv. “Cross-view” – two units arranged to maximize overlapping active volume to get multiple perspectives on the same scene. The two distinct coordinate systems for each unit would be harmonized into a single world coordinate system through mutual calibration.



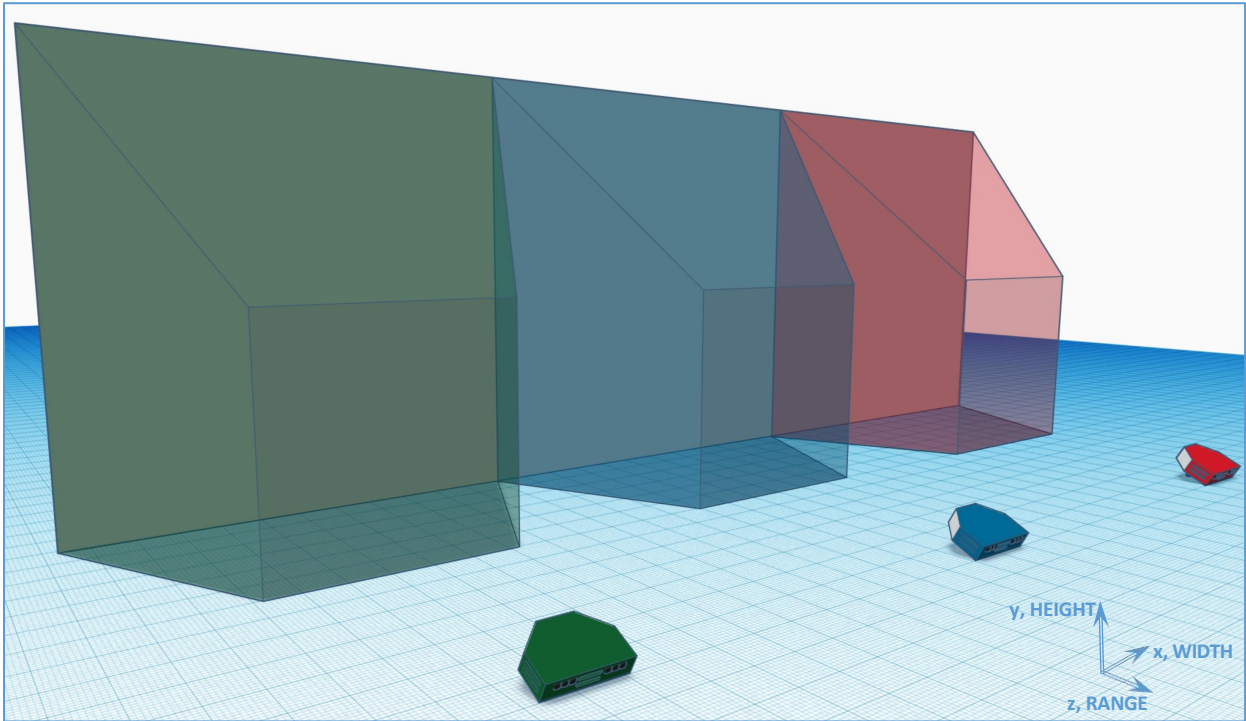
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B. MINI configuration; for active range dimensions, refer to Figure 1B. PFDM units shown at ground level for illustration purposes only; units can be positioned at any height to optimize vantage point onto experimental stage.

i. Single unit, shown for reference

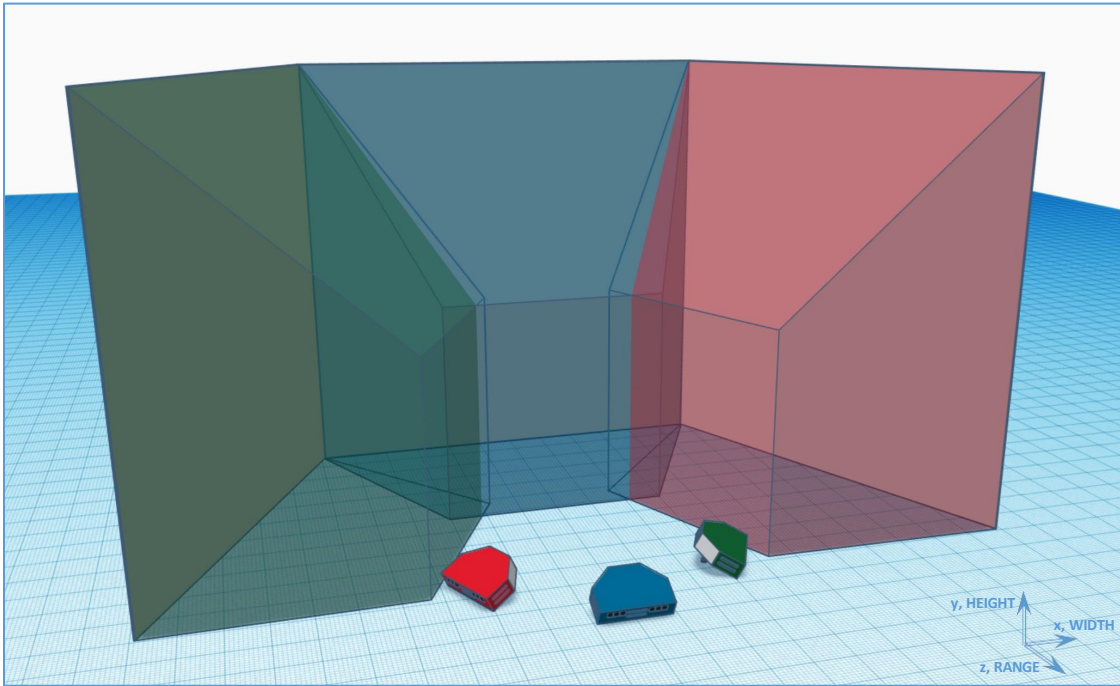


ii. “Array” – multiple units arranged in a linear array for viewing adjacent scenes concurrently

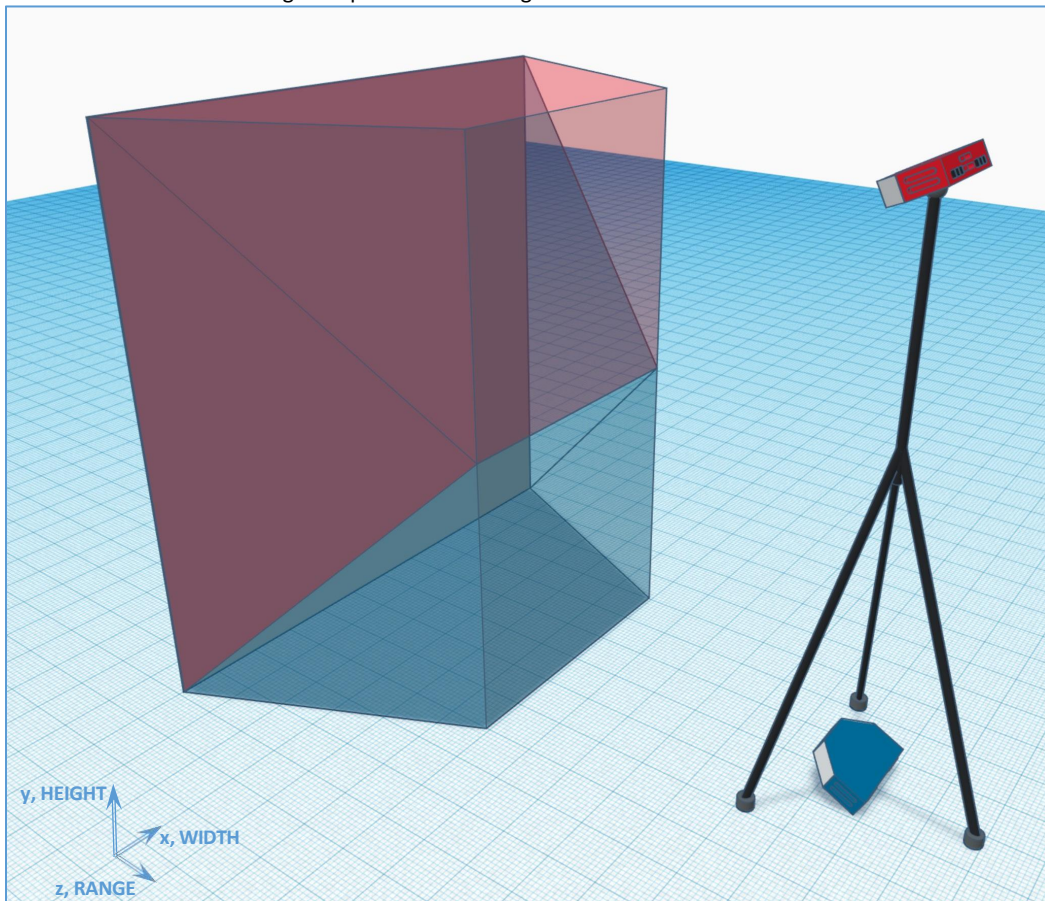


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- iii. “Fan” – multiple units arranged to maximize contiguous active volume. Some overlap of active regions may result. Unit height may need to be staggered (not shown) to avoid optical obscurations.



- iv. “Stack” – two units arranged to produce a full-height active volume.



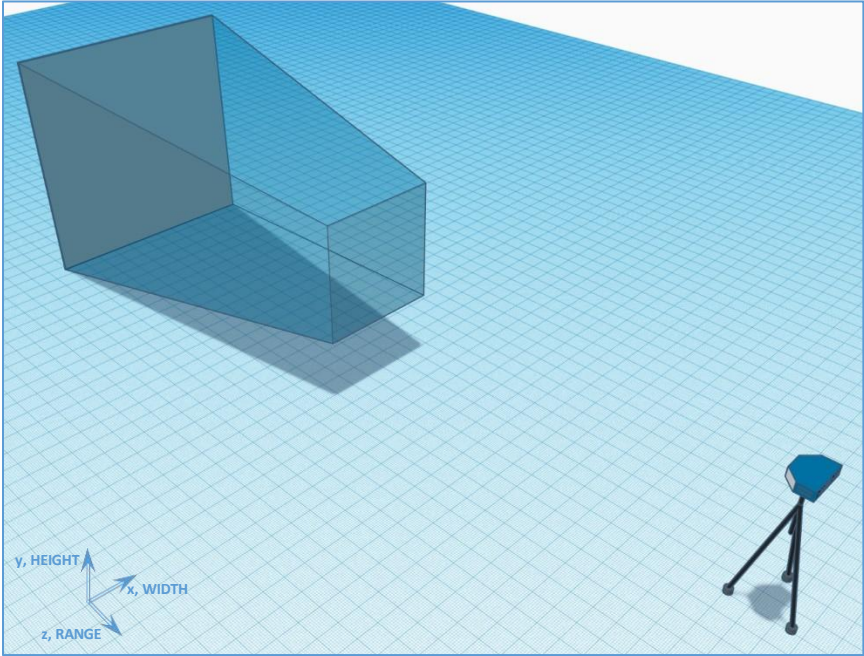
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APPENDIX 2

Example of multiple regions of interest

A. LAB configuration; for active range dimensions, refer to Figure 1A. PFDM units shown on tripod at 1.5m above ground level for illustration purposes only; units can be positioned at any height to optimize vantage point onto experimental stage.

i. Single full-scale region of interest, shown for reference



ii. Two sectional regions of interest; user specifies region(s) of interest at the time of experiment initiation, or defaults to single full-scale active region. Specifying regions of interest allows the user to intentionally exclude unwanted visual data (e.g., humans, goats, trees, etc.) and minimize data storage size to allow for longer experimental runs.

